

CAD-Criminal Apprehension Drone

Project Report submitted for

Design Thinking and Innovation (24UC1203)

A.Y. 2024-25, Odd Semester

By Team K

DTI Section No: **10**

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DECLARATION

I, **Avinash Yarukala**, hereby declare that the project report titled "**CAD-Criminal Apprehension Drone**" submitted by me for the Design Thinking and Innovation (24UC1203) course, under the supervision of **Dr. J. Pundareekam Goud** and **Dr. Srinivas Anga**, is my original work. The report is prepared based on my own research and analysis.

I further declare that:

1. The contents of this project report have not been submitted, in whole or in part, for any other degree or diploma.
2. The project work has not been previously published or used by anyone else.
3. Any assistance received in preparing this report, either technical or otherwise, has been duly acknowledged.
4. All data, figures, and diagrams included in this report are authentic and have been sourced properly.

I understand that any attempt to present another person's work as my own will be considered plagiarism and may lead to disciplinary action as per the institution's regulations.

Date: 04/12/2024

Place: Bachupally.

Team Leader Signature: _____

DTI Trainer's Endorsement:

I certify that I have supervised the preparation of this project report and that to the best of my knowledge, the above-mentioned statements made by the student are correct.

DTI Trainer-1 Signature: _____

Name: Dr. J. Pundareekam Goud

Date:

DTI Trainer-2 Name: Dr. Srinivas Anga

Signature: _____

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Project Description

The proliferation of drones has transformed various industries, offering innovative solutions for aerial photography, delivery services, and surveillance. However, this rapid advancement has also given rise to significant challenges, particularly in the realm of criminal activities. Criminal organizations have begun to exploit drone technology for nefarious purposes, ranging from drug smuggling to reconnaissance for illegal operations. The ability of drones to operate discreetly, often at low altitudes, allows them to evade traditional detection methods, making them an appealing tool for criminals.

In recent years, law enforcement agencies have reported a surge in incidents involving drones being used for illegal activities. These incidents include delivering contraband into prisons, facilitating human trafficking, and even conducting targeted attacks. The versatility and relatively low cost of consumer-grade drones have made them accessible to a broader range of potential offenders, further complicating efforts to regulate and monitor their use.

Moreover, the integration of advanced technologies, such as artificial intelligence and machine learning, enhances the capabilities of these drones, allowing for autonomous operations and sophisticated data gathering. This evolution poses a challenge for policymakers and law enforcement, who must develop strategies to counteract the misuse of drones while balancing the rights of legitimate users.

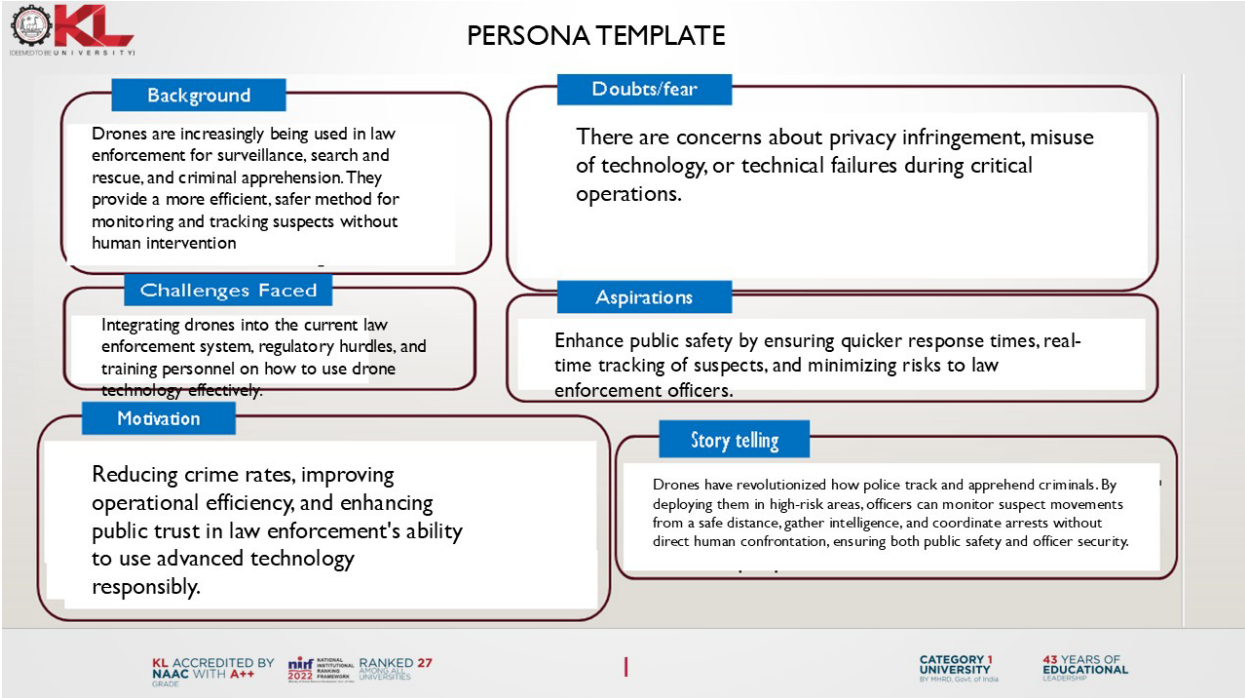
Addressing the issue of criminal drone usage requires a multi-faceted approach that includes regulatory measures, technological innovations for detection and prevention, and comprehensive training for law enforcement personnel to identify and respond to drone-related threats effectively.

Design Thinking Process

2.1 Empathize----- 2.1.1. Interviewing



2.1.2. Persona



2.2.Define

2.2.1.Custom Journey Map: Criminal Apprehension Drone

This journey map outlines the key stages in the operation of a criminal apprehension drone, from deployment to capture, focusing on user interactions, drone functionality, and stakeholder needs.

Stage	User Actions	Drone Activities	Touchpoints	Pain Points	Opportunities
1. Incident Alert	Law enforcement receives a crime alert	Standby, awaiting deployment signal	Crime detection systems, Police dashboard	Delays in receiving alerts	Automate alert triggers
2. Deployment	Officer deploys drone via app or control	Drone takes off and sets course	Mobile app, Control station	Complex UI for deployment	Simplify deployment interface
3. Surveillance	Monitors drone feed for updates	Real-time video feed, thermal imaging	Live video stream	Poor connectivity	Integrate 5G, local caching
4. Tracking	Identifies suspects via drone visuals	Facial recognition, object tracking	AI analytics, Body cams	Accuracy of AI recognition	Improved AI model training
5. Communication	Coordinates with ground units	Drone relays position data to units	Radio, App notifications	Lag in information transfer	Direct integration with radios
6. Apprehension	Directs officers to location	Throws nets, spotlights on suspects	Police GPS, Maps integration	Miscommunication in directions	Auto-routing to closest units
7. Return & Recharge	Confirms safe return to base	Navigates to home base, recharges	Drone dock, Energy systems	Battery life	Fast-charging, swappable batteries

2.2.2.Point of View Statements for a Criminal Apprehension Drone:

1. **For Law Enforcement Officers:**

Officers need a reliable tool that provides real-time situational awareness and supports safe, efficient apprehensions without adding to their operational burden.

2. **For Drone Operators:**

Operators require a user-friendly interface and robust communication systems to maintain control and adapt quickly in dynamic, high-stress environments.

3. **For the Community:**

Community members need assurance that the drone enhances public safety while respecting privacy, civil rights, and fostering trust through transparency and accountability

2.2.3.Problem Statement

Problem Statement

Hyderabad faces a challenge in ensuring timely response to crimes and effective identification of suspects. Addressing these challenges is crucial for realizing the full potential of drones in enhancing public safety and law enforcement efforts in Hyderabad.

Timely response to crimes



Effective identification of suspects



Public concerns about privacy and surveillance



Technical limitations in urban environments



Complex legal and regulatory frameworks



2.3.Ideate

2.3.1.Brainstorming (Idea generation)

Tear Gas Deployment in Extreme Cases – 2 points	Ashwin
Drones Equipped with Sirens to Alert Civilians – 7 points	Ashwin
Live-Streaming to Command Center – 9 points	Ashwin
Drone-Assisted Search and Rescue for Hidden Suspects – 8 points	Ashwin
Predefined Patrol Routes for Drones in High-Risk Areas – 6 points	Ashwin
Thermal Imaging for Night Pursuits – 9 points	Avinash
Real-Time Tracking of Suspects – 8 points	Avinash
Facial Recognition for Identifying Known Criminals – 10 points	Avinash
Automatic License Plate Recognition – 6 points	Avinash
Drone Swarm for Crowd Surveillance – 5 points	Avinash
Thermal Scanning in Dense Foliage – 8 points	Charan tej
Geo-Fencing to Limit Suspect Movement – 5 points	Charan tej
Drone Strobes to Deter Flight or Concealment – 3 points	Charan tej
In-Air Charging Stations for Continuous Surveillance – 4 points	Charan tej
Using Drones as Witnesses in Court Cases (Video Evidence) – 9 points	Charan tej
Remote Control Handover to Nearby Units – 5 points	Hitesh
Suspect Signal Jammer for Isolated Areas – 4 points	Hitesh
Automatic Detection of Unusual Movements – 7 points	Hitesh
Instant Replay of Surveillance Footage for Officers – 6 points	Hitesh
Assistance in Bomb Disposal Situations – 5 points	Hitesh
Artificial Intelligence for Predictive Patrols – 7 points	Mani charan chava
Obstacle Avoidance for Urban Tracking – 6 points	Mani charan chava
Drone Surveillance in Drug Trafficking Hotspots – 7 points	Mani charan chava
Tracking Vehicles in High-Speed Chases – 8 points	Mani charan chava
Drone-Aided Evidence Collection – 6 points	Mani charan chava
Remote Communication with Suspects via Drone – 4 points	Ram
Emergency Medical Aid Drops – 3 points	Ram
GPS Tagging Suspect Vehicles – 9 points	Ram
Spotlight for Illuminating Suspects at Night – 6 points	Ram
Audio Warnings or Commands to Suspects – 5 points	Ram
Drone-Based Voice Mimicry for Confusing Suspects – 3 points	Rishitha
Drones for Enhanced Visibility in Low-Weather Conditions – 8 points	Rishitha
Integration with Law Enforcement Databases for Real-Time Info – 8 points	Rishitha
Detachable Drone Camera for Evidence in Small Spaces – 5 points	Rishitha
Real-Time Translation for Commands to Non-English Speakers – 4 points	Rishitha

Enhanced Audio Surveillance in Noisy Environments – 6 points	Stuthi
Multi-Drone Coordination for Surrounding Suspects – 7 points	Stuthi
Drones for Surveillance of Known Crime Locations – 7 points	Stuthi
Augmented Reality Interface for Tracking Multiple Drones – 4 points	Stuthi
Infrared Scanning for Detecting Concealed Weapons – 8 points	Stuthi

2.3.2. Idea Evaluation (Grouping, multi voting, Rapid Estimation Forms)

Facial Recognition for Identifying Known Criminals – 10 points
Using Drones as Witnesses in Court Cases (Video Evidence) – 9 points
Thermal Imaging for Night Pursuits – 9 points
GPS Tagging Suspect Vehicles – 9 points
Live-Streaming to Command Center – 9 points
Real-Time Tracking of Suspects – 8 points
Drone-Assisted Search and Rescue for Hidden Suspects – 8 points
Tracking Vehicles in High-Speed Chases – 8 points
Infrared Scanning for Detecting Concealed Weapons – 8 points
Integration with Law Enforcement Databases for Real-Time Info – 8 points

AFTER VOTING IT TO 100 POINTS WEIGHTAGE



Facial Recognition for Identifying Known Criminals – 28 points
Using Drones as Witnesses in Court Cases (Video Evidence) – 24 points
GPS Tagging Suspect Vehicles – 18 points
Real-Time Tracking of Suspects – 16 points
Infrared Scanning for Detecting Concealed Weapons – 14 points

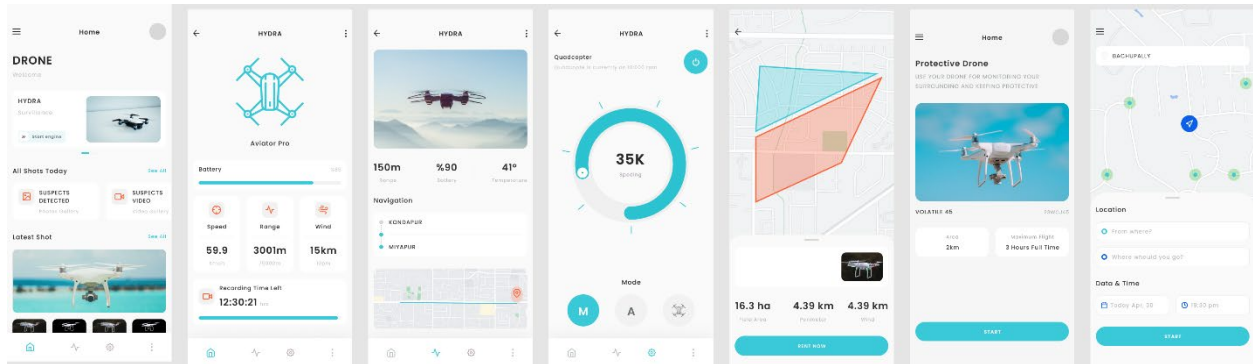
AFTER VOTING IT TO 1000 WEIGHTAGE



Facial Recognition for Identifying Known Criminals – 450 points
GPS Tagging Suspect Vehicles – 400 points
Real-Time Tracking of Suspects – 100 points
Infrared Scanning for Detecting Concealed Weapons – 50 points

2.4.Prototype

2.4.1.Low Fidelity Prototypes



FORM 2 THE PATENTS ACT, 1970 (39 of 1970) & The Patent Rules, 2003 COMPLETE SPECIFICATION (See sections 10 & rule 13)		
1. TITLE OF THE INVENTION C.A.D-Criminal Apprehension Drone		
2. APPLICANT (S)		
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Title

(Please Fill – One Sentence)

Technical Field

This invention pertains to the law enforcement and public safety sector, specifically to advanced technologies for crime prevention and criminal apprehension. It is designed for deployment by police departments, defense forces, and government agencies.

(Please fill – Applicable Area of invention)

Technical

Field

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Background Of The Invention

Traditional law enforcement responses to crimes such as burglaries or vehicle pursuits are often delayed due to logistical constraints. This time lag can enable criminals to evade capture, posing a significant challenge to maintaining law and order. Existing technologies lack the adaptability and immediacy required to address these situations effectively.

The Criminal Apprehension Drone (C.A.D.) addresses these challenges by providing a rapid, automated, and versatile solution. This invention integrates cutting-edge tools such as non-lethal tasers, net deployment systems, and vehicle tracking mechanisms, offering a dynamic approach to crime prevention and resolution.

(Please fill – Provide details of similar inventions (4-5 Nos.))

(The "Background of the Invention" section in an Indian utility patent serves as a narrative that provides context and background information about the problem or need that the invention aims to address.

In this section, the inventor provides a detailed explanation of the state of the art, existing technologies, and any challenges or shortcomings associated with current solutions. The Background of the Invention helps to set the stage for the reader,

providing a clear understanding of why the invention is necessary and what problems or deficiencies it seeks to overcome.)

Objects Of The Invention

1. To develop a drone-based system for immediate deployment in criminal incidents, enhancing response efficiency for law enforcement.
2. To incorporate multiple modular attachments for versatile use, including non-lethal apprehension tools (e.g., taser, net launcher) and vehicle tracking devices.
3. To establish a scalable and networked system of drones capable of creating a proactive and invisible security web for public safety.

(Please fill – 2-3 Objectives in description manner)

(The "Objects of the Invention" section in an Indian utility patent outlines the specific goals and objectives that the invention aims to achieve.

In this section, the inventor articulates the specific problems or needs identified in the Background of the Invention and describes how the invention overcomes these challenges or deficiencies. The Objects of the Invention help to clarify the intended benefits and the practical applications of the patented technology.)

Summary

The Criminal Apprehension Drone (C.A.D.) is an innovative unmanned aerial system designed to assist law enforcement in real-time crime prevention and suspect apprehension. The drone is equipped with a modular attachment system, enabling it to deploy a net or a non-lethal taser to subdue individuals attempting to flee. In cases where the suspect escapes via vehicle, the drone can launch a magnetic AirTag-like tracking device to monitor the vehicle's movements.

The C.A.D. system is envisioned as a scalable security solution, enabling multiple drones to form a coordinated network. This creates a dynamic, responsive, and invisible web of security that can be adapted to various operational needs, ranging from urban crime prevention to border defense. By leveraging this technology, governments and defense forces can enhance public safety and maintain law and order more effectively.

(Please fill – One Paragraph about the invention)

Summary

The Criminal Apprehension Drone (C.A.D.) is an innovative unmanned aerial

system designed to assist law enforcement in real-time crime prevention and suspect apprehension. The drone is equipped with a modular attachment system, enabling it to deploy a net or a non-lethal taser to subdue individuals attempting to flee. In cases where the suspect escapes via vehicle, the drone can launch a magnetic AirTag-like tracking device to monitor the vehicle's movements.

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BRIEF DESCRIPTION OF THE DRAWINGS

(Please fill – All the figures with Caption)

(In an Indian utility patent, the "Brief Description of the Drawings" is a section that accompanies the drawings or diagrams included in the patent document. This section provides a written explanation or narrative that helps readers understand the content and significance of each drawing or figure. It is important because it adds context and clarifies the details presented visually in the drawings.

BRIEF DESCRIPTION OF THE TABLES (IF REQUIRED)

(Please fill – All the tables with Caption)

(In an Indian utility patent, the "Brief Description of the Tables" is a section that accompanies any tables included in the patent document. This section provides a written explanation or narrative that helps readers understand the content and significance of each table. Similar to the "Brief Description of the Drawings," this section adds context and clarifies the information presented in tabular form.

Detailed Description

The Criminal Apprehension Drone (C.A.D.) is a quadcopter designed for rapid deployment in high-pressure scenarios. Its key features include:

1. Modular Attachments:

- Net Launcher: A system designed to immobilize suspects by deploying a strong

yet lightweight net.

- Non-Lethal Taser: Provides a non-lethal method to incapacitate suspects.
- Magnetic Vehicle Tracker: A puck-like device equipped with GPS tracking, designed to magnetically attach to fleeing vehicles.

2. Dynamic Adaptability:

The drone's modular system allows it to be equipped with different tools depending on the situation, making it suitable for diverse operations, including crime prevention, border surveillance, and urban security.

3. Networked Security System:

Multiple C.A.D. units can be deployed to work collaboratively, creating a comprehensive and coordinated security network. This enhances coverage and ensures swift responses to incidents across larger areas.

4. Wide Applicability:

The drone is envisioned as a versatile solution for police departments, defense forces, and government agencies, enabling them to establish a crime-free security network across urban and rural regions.

(Please fill – Provide details on invention – purpose, methodology, testing details (if any))

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(Please fill – What kind of Technique/Method/Flow-process/technology you want to protect in your invention)

Claims

1. A drone-based system equipped with modular tools for non-lethal apprehension, including a net launcher, taser, and vehicle tracking device.
2. A networked drone system capable of collaborative operations, forming a scalable and proactive security framework.
3. A magnetic vehicle tracking device designed to enhance suspect monitoring and apprehension.

ABSTRACT

(Please fill – Write in brief about your invention – 250 words)

Abstract

The Criminal Apprehension Drone (C.A.D.) is an innovative solution for real-time criminal apprehension and public safety. Equipped with modular tools such as a net launcher, non-lethal taser, and magnetic vehicle tracker, this drone provides versatile and rapid response capabilities for law enforcement. Designed for deployment by police, defense forces, and government agencies, the C.A.D. system can form a coordinated security network, creating an adaptable and scalable framework to prevent and resolve crimes.

Learn Canvas

Designed for:	Designed by:	Date:	Version:
Criminal apprehension	Team K	06/11/2024	0.1

Problem 1) Inefficiency in tracking and locating suspects in real-time. 2) High risk to law enforcement personnel in direct pursuits. 3) Limited reach and slow response in rural or hard-to-access areas.	Solution 1) Drones equipped with high-definition cameras and thermal imaging for enhanced surveillance. 2) Real-time data transmission to law enforcement for quick decision-making. 3) Autonomous flight paths and GPS tracking for fast deployment.	Unique Value Proposition Real-time, safe, and efficient suspect tracking and apprehension through aerial drone surveillance, reducing risks to personnel and enabling coverage in difficult terrains	Unfair Advantage Proprietary technology and algorithms for advanced imaging and tracking; partnerships with law enforcement agencies; secure data transmission protocols that competitors can't easily replicate.	Customer Segments 1) Law enforcement agencies 2) Government security departments 3) Private security firms
Existing Alternatives 1) Helicopters for aerial surveillance 2) Ground-based tracking teams 3) CCTV surveillance in urban areas	Key Metrics 1) Number of suspects tracked and apprehended 2) Reduction in response time 3) Successful missions in hard-to-access locations 4) Operational cost savings compared to traditional methods	High-Level Concept "Uber for Criminal Apprehension" – drones dispatched to track suspects like ride-hailing services dispatch cars.	Channels 1) Direct partnerships with law enforcement agencies 2) Government contracts 3) Security industry conferences and trade shows	Early Adopters Law enforcement agencies in urban areas with high crime rates, rural areas with limited resources, and private security firms needing advanced surveillance.

Cost Structure 1) Drone equipment and maintenance 2) R&D for proprietary tracking algorithms 3) Data storage and transmission costs	Revenue Structure 1) Contract-based revenue with law enforcement and government agencies 2) Subscription model for private security firms 3) Fee-for-service model for specific missions
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